

Ex 1

Assignment 3: $\phi > 0$

$$\frac{d}{d\phi} \int_{2\phi}^{4\phi} f(x, \phi) dx \quad \text{with} \quad f(x, \phi) = \begin{cases} x/\phi - 1, & x < 3\phi \\ 1, & x \geq 3\phi \end{cases}$$

$$\begin{aligned} \int_{\text{int}} &= \int_{2\phi}^{3\phi} \frac{d(x/\phi - 1)}{d\phi} dx + \int_{3\phi}^{4\phi} \frac{d}{d\phi} dx \\ &= \int_{2\phi}^{3\phi} \frac{-x}{\phi^2} dx + 0 = \left. \frac{-x^2}{2\phi^2} \right|_{2\phi}^{3\phi} = \frac{(-9+4)\phi^2}{2\phi^2} = \boxed{\frac{-5}{2}} \end{aligned}$$

$$\begin{aligned} \int_{\text{limit}} &= f_{\text{upper}} - f_{\text{lower}} = f(4\phi, \phi) \cdot \frac{d(4\phi)}{d\phi} - f(2\phi, \phi) \cdot \frac{d(2\phi)}{d\phi} \\ &= 1 \cdot 4 - (2-1) \cdot 2 = \boxed{2} \end{aligned}$$

$$\begin{aligned} \int_{\text{edge}} &= -\left(f^+(3\phi, \phi) - f^-(3\phi, \phi)\right) \frac{d(3\phi)}{d\phi} \\ &= -3(1 - 2) = \boxed{3} \end{aligned}$$

$$\Rightarrow \frac{d}{d\phi} \int_{2\phi}^{4\phi} f(x, \phi) dx = \frac{-5}{2} + 2 + 3 = \boxed{\frac{5}{2}} \quad \checkmark$$